DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I

COMPILER DESIGN

Class: B. E (CSE)
Semester: VI
Academic Year: 2025-2026

Max Marks: 30
Duration: 1hr 30 min
Date: 25-03-2026

Q.NO	PART A Answer All Questions (5 x 2 = 10 M)	Marks	BTL	CO
1	What is Left factoring? Explain with an example.	2	2	1
2	Write the procedure to eliminate left recursion and apply on the given Grammar $E \rightarrow E + E / E * E / id$	2	3	1
3	What is Cross compiler? Give an example.	2	1	5
4	When do we say a Grammar is Ambiguous? Explain with an example.	2	4	1
5	What is Symbol Table and What are the operations performed on Symbol Table.	2	2	4
	PART B **Answer Any Two Questions (2*10 = 20 Marks)**			
6	a) Explain the phases of compiler in detail with a neat diagram.	5	2	5
	b) Explain how regular expressions are used in LEX to define tokens. Demonstrate with a sample LEX program that identifies identifiers, keywords, and numbers.	5	1	5
7	a) Mention the rules for finding First () and Follow () sets for the grammar. $E \rightarrow E + T / T$ $T \rightarrow T * F / F$ $F \rightarrow (E) / a / b$	5	3	2
	b) Explain roles and responsibilities of Symbol table and Error handler?	5	2	4
8	a) Explain the Block Diagram of LL (1) Parser with a neat Diagram?	5	2	2
	b) Specify the functionalities of a) Linker b) Loader c) Assembler d) Preprocessor	5	2	2